



Propagation of arbitrary initial wave packets in a quantum parametric oscillator: Instability zones for higher order moments

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ABSTRACT

We consider the dynamics of a particle in a parametric oscillator with a view to exploring any quantum feature of the initial wave packet that shows divergent (in time) behaviour for parameter values where the classical motion dynamics of the mean position is bounded. We use Ehrenfest's theorem to explore the dynamics of n th order moment which reduces exactly to a linear non autonomous differential equation of order $n+1$. It is found that while the width and skewness of the packet is unbounded exactly in the zones where the classical motion is unbounded, the kurtosis of an initially non-gaussian wave packet can become infinitely large in certain additional zones. This implies that the shape of the wave packet can change drastically with time in these zones.

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The quantum dynamics in parametric oscillators (a model for dynamics of ions in a Paul trap) has produced a lot of interesting results from both analytical [1–6] and computational [8–10] standpoints. In all these studies, the initial wave packet has been taken to be the generic wave packet which is a *Gaussian*. The *Gaussian* wave packet has the advantage that it remains a *Gaussian* as time elapses. This allows the study of complicated situations like the rotating wave packet which was studied recently [11]. In this communication, we want to focus on initial wave functions $\Psi_0(x)$ which are arbitrary and point out certain general features of initial wave packets. Specifically, we consider the one dimensional Hamiltonian.

$$\mathcal{H} = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2(1 + \epsilon f(t)) \quad (1)$$

where $f(t) = f(t + \frac{2\pi}{\Omega}) = f(t + T)$ is a function with periodicity (T). For most of the discussion we will take $f(t) = \cos(\Omega t)$. At any time t the wave function $\Psi_0(x, t)$ has the form $\sqrt{P(x, t)} \exp[i\phi(x, t)]$, where $P(x, t)$ is the time dependent probability distribution. We shall focus here on the mean position $\langle x \rangle$, the variance $\Delta^2 = \langle x^2 \rangle - \langle x \rangle^2$, the skewness $S = \langle (x - \langle x \rangle)^3 \rangle$ and the kurtosis $K = \langle (x - \langle x \rangle)^4 \rangle$, which are the first

few moments of the probability distributions. As for the phase we will restrict our discussion to the first two terms in a *Taylor expansion* of $\phi(x, t)$ around $x = 0$, i.e. $\phi(x, t) = k(t)x + \frac{\gamma(t)}{2}x^2 + \dots$. We will make use of Ehrenfest's theorem for the moments and the phase which allow the study of dependence on initial conditions, as seen for the Gross-Pitaevskii equation for both single [12] and binary [13] condensates. The solution of $\Psi(x, t)$ for Eq. (1) has been obtained in the most comprehensive form by authors of [4] and [5] as discussed in detail in the review of [6]. In [4] the technique used is path integrals and in [5] the same answer is obtained by exploiting the analogy between the quantum mechanical equation of motion for the operator x and the classical Mathieu equation. It has been clearly noted by Brown [4] that the solution is valid only when the classical motion is bounded. Similarly, as noted in the review of Leibfried et al. [6], the solution of Glauber [5] uses the Floquet form of the solution of Mathieu equation which is valid when the solution is bounded. In ref. [4], a canonical transformation on the Hamiltonian $\mathcal{H}$ given in Eq. (1) by the unitary operator $\mathcal{U} = \exp[-i\chi(t)x^2(t)]$ with $\chi = \frac{m}{4}[\frac{\dot{f}}{f} + \frac{f^*}{f}]$ is carried out to reduce the Hamiltonian to a quadratic form. A subsequent transformation to remove the cross term results in a standard simple harmonic oscillator with an overall time dependent scale factor. A similar result had been previously obtained [7] for the very special case of $f(t) = \sum_n \delta(t - nT)$. The phase transition in the quantum dynamics described over there is an immediate consequence of our Eq. (1) above when $f(t)$ is a sequence of delta functions. It has been carefully noted in refs. [4–6] that

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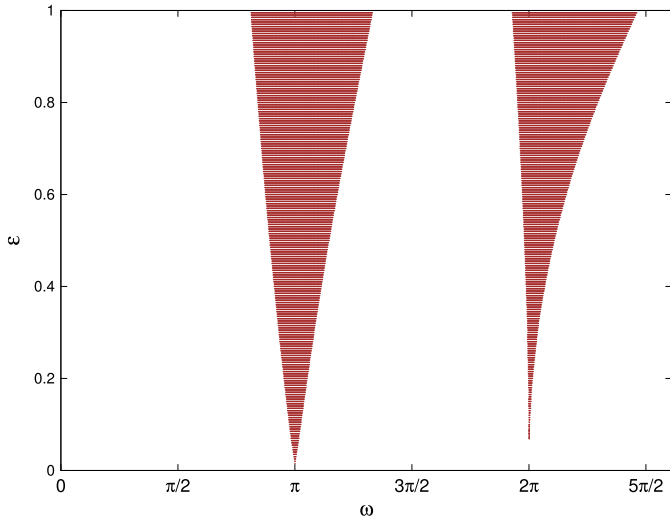


Fig. 1. Zones of bounded and unbounded solutions for $\langle x \rangle$. The brown shaded regions show the unstable zones. (For interpretation of the colours in the figure(s), the reader is referred to the web version of this article.)

all the consequences of the above transformations and the corresponding simplifications are relevant in the stability zone. Our interest in this work lies only in the instability zones. The usual analytic study of instability is restricted to the dynamics of the average position. It is the instability zone of the higher moments (variance, skewness and kurtosis) that we are interested in. We find that for studying the zone of divergent solutions, Ehrenfest's theorem is an ideal starting point.

It is known that the mean position $\langle x \rangle$ of the particle follows the Mathieu equation

$$\frac{d^2}{dt^2} \langle x \rangle + \omega^2 (1 + \epsilon f(t)) \langle x \rangle = 0 \quad (2)$$

whose properties are well known. All initial wave packets will have run-away (diverging exponentially with time) solution within certain bands (as shown in Fig. 1).

The bands originate from the points $\omega = \frac{\Omega}{2}$ (primary resonance with response period $2T$), $\frac{3\Omega}{2}$, $\frac{5\Omega}{2}$... etc. and from $\omega = \Omega$ (secondary resonance with response period T), 2Ω , 3Ω ... etc. if we imagine varying ω for a fixed modulation frequency Ω . From the known [14] properties of instability zones of the Mathieu equation, the band near $\frac{\Omega}{2}$ is bounded by the curves $\omega_{1,2}(\epsilon) = \frac{\Omega}{2} + \Delta\omega_{1,2}(\epsilon)$ with $\Delta\omega_1 = -\frac{\epsilon\Omega}{8} - \frac{\epsilon^2\Omega}{128}$ and $\Delta\omega_2 = \frac{\epsilon\Omega}{8} - \frac{\epsilon^2\Omega}{128}$ giving a width of $\mathcal{O}(\epsilon^2)$. For the first secondary resonance the boundary curves are $\omega_{1,2}(\epsilon) = \Omega + \Delta\omega'_{1,2}(\epsilon)$ with $\Delta\omega'_1 = -\frac{\epsilon^2\Omega}{24} + \mathcal{O}(\epsilon)^3$ and $\Delta\omega'_2 = \frac{5\epsilon^2\Omega}{24} + \mathcal{O}(\epsilon)^3$ giving a width of $\mathcal{O}(\epsilon^2)$ which is much smaller than the width of the primary resonance for $\epsilon \ll 1$.

We find the following

- A) An initial Gaussian wave function remains a Gaussian within our study of the different order moments as it should.
- B) The variance Δ^2 satisfies a third order equation with periodic coefficients (this equation was first noted by Hänggi and Zerbe [15]). The resonance of width $\mathcal{O}(\epsilon)$ for this equation should have been at $\omega = \frac{\Omega}{4}$ as inferred from a combination of Floquet theory [14] and harmonic balance [14]. This has never been found in any numerical work. The most intensive numerical investigation has been carried out by Hashemloo et al. [9] and all their results are in agreement with what we find from the analysis of the dynamical system for Δ^2 . We provide an explanation (initial wave packet independent)

of why this resonance zone is not supported. More importantly, we find the secondary resonance in this case is of width $\mathcal{O}(\epsilon)$ and actually matches the width of the primary resonance for $\langle x \rangle$. Consequently, for no initial wave packet will there be a region in (ω, ϵ) space where the width diverges but the average position is bounded.

- C) The skewness factor S satisfies a fourth order differential equation with periodic coefficients. Now the expected primary resonances of width $\mathcal{O}(\epsilon)$ are at $\omega = \frac{\Omega}{6}$ and $\frac{\Omega}{2}$. The resonance at $\frac{\Omega}{6}$ is not seen but the resonance at $\frac{\Omega}{2}$ exists and has the same width as the resonance in $\langle x \rangle$ and Δ^2 . Within this width an initial skewness is greatly magnified as time elapses. It is also seen that if the wave packet is initially symmetric ($S = 0$) about the average value, it remains symmetric for ever.
- D) The kurtosis K satisfies a fifth order differential equation with periodic coefficients. There are two expected primary resonances (width $\mathcal{O}(\epsilon)$) at $\omega = \frac{\Omega}{8}$ and $\omega = \frac{\Omega}{4}$. In this case, we find the kurtosis increases indefinitely in a band of width $\mathcal{O}(\epsilon)$ around both $\frac{\Omega}{8}$ and $\frac{\Omega}{4}$. Interestingly enough, there is also a very wide resonance zone around $\frac{\Omega}{2}$. The indefinite increase in K implies a qualitative change in the tail of the distribution.

We first establish the equations of motion of the higher order moments. This follows from repeated use of Ehrenfest's theorem for the expectation value $\langle \mathcal{O} \rangle$ of any operator $\mathcal{O}$, which is written as

$$i\hbar \langle \dot{\mathcal{O}} \rangle = \langle [\mathcal{O}, H] \rangle + i\hbar \frac{\partial}{\partial t} \langle \mathcal{O} \rangle \quad (3)$$

The last term on the right hand side is necessary only if the operator carries an explicit time dependence. Using the $\mathcal{H}$ of Eq. (1), we see immediately that $\langle \dot{x} \rangle = \frac{\langle p \rangle}{m}$ and $\langle \dot{p} \rangle = -m\omega^2(1 + \epsilon f)\langle x \rangle$, leading to Eq. (2). For Δ^2 , we need the following time derivatives calculated from Eq. (3)

$$\begin{aligned} \frac{d}{dt} \langle x^2 \rangle &= \frac{\langle xp + px \rangle}{m} \\ \frac{d}{dt} \langle p^2 \rangle &= -m\omega^2(1 + \epsilon f(t)) \langle xp + px \rangle \end{aligned} \quad (4)$$

$$\frac{d}{dt} \langle xp + px \rangle = 2 \frac{\langle p^2 \rangle}{m^2} - 2\omega^2(1 + \epsilon f(t)) \langle x^2 \rangle$$

Suitable derivatives and eliminations yield

$$\frac{d^3}{dt^3} \langle x^2 \rangle + 4\omega^2(1 + \epsilon f(t)) \frac{d}{dt} \langle x^2 \rangle + 2\omega^2 \epsilon \dot{f} \langle x^2 \rangle = 0 \quad (5)$$

As for $\langle x \rangle^2$, $\frac{d}{dt} \langle x \rangle^2 = 2 \frac{\langle x \rangle \langle p \rangle}{m}$ and $\frac{d}{dt} \langle p \rangle^2 = -2m\omega^2(1 + \epsilon f(t)) \langle x \rangle \langle p \rangle$ which ensure that $\langle x \rangle^2$ satisfies an equation identical to Eq. (5) and hence

$$\frac{d^3}{dt^3} \Delta^2 + 4\omega^2(1 + \epsilon f(t)) \frac{d}{dt} \Delta^2 + 2\omega^2 \epsilon \dot{f} \Delta^2 = 0 \quad (6)$$

Repeated derivatives and algebraic manipulations similar to those used for obtaining Eq. (6), lead to

$$\begin{aligned} \frac{d^4}{dt^4} S + 10\omega^2(1 + \epsilon f(t)) \frac{d^2}{dt^2} S + 9\omega^4(1 + \epsilon f(t))^2 S \\ + 10\omega^2 \epsilon \dot{f} \frac{d}{dt} S + 3\omega^2 \epsilon \ddot{f} S = 0 \end{aligned} \quad (7)$$

and

$$\begin{aligned} \frac{d^5}{dt^5} K + 20\omega^2(1 + \epsilon f(t)) \frac{d^3}{dt^3} K + 24\omega^2 \epsilon \dot{f} \frac{d^2}{dt^2} K \\ + 64\omega^4(1 + \epsilon f(t))^2 \frac{d}{dt} K + 18\omega^2 \epsilon \ddot{f} \frac{d}{dt} K \\ + 64\omega^4 \epsilon (1 + \epsilon f(t)) \dot{f} K + 4\omega^2 \epsilon \ddot{f} K = 0 \end{aligned} \quad (8)$$

for the skewness and kurtosis respectively. The above equations are exact. What Eqs. (6)–(8) tell us is, given an arbitrary initial wave function with width Δ_0 , skewness S_0 and kurtosis K_0 , we can determine how Δ , S and K develop with time. This gives us some idea of the time dependence of the probability distribution itself. It should be noted that the n th order moment leads to a $(n+1)^{th}$ order differential equation for the process. The function $k(t)$ and $\gamma(t)$ characterizing the phase are obtained from the expectation value $\hbar k(t) = \langle p \rangle$ and $2\gamma(t)\hbar = \frac{d}{dt} \ln(\Delta^2)$.

We begin with the analysis of Eq. (6) with $f(t) = \cos(\Omega t)$. For $\epsilon = 0$, we have the zeroth order solution

$$\Delta_0^2 = A_0 + A_1 \cos(2\omega t) + B_1 \sin(2\omega t) \quad (9)$$

which is the familiar result for the dynamics of a simple harmonic oscillator. We seek the periodic solution of Eq. (6) when $\epsilon \neq 0$. In analogy with the primary resonance of Eq. (2), where the basic solution is a linear combination of $\cos(\omega t)$ and $\sin(\omega t)$, we expect the primary resonance of Eq. (6) to be around $\omega = \frac{\Omega}{4}$. Thus for $\epsilon \neq 0$, we seek the boundaries $\omega = \frac{\Omega}{4} + \delta(\epsilon)$, where the solution of Eq. (9) is periodic. Using the perturbation theory with $\delta(\epsilon) = \delta_1\epsilon + \delta_2\epsilon^2 + \delta_3\epsilon^3 + \dots$ (where $\delta_1, \delta_2, \delta_3, \dots$ are coefficient of $\mathcal{O}(1)$) and $\Delta^2 = \Delta_0^2 + \epsilon\Delta_1^2 + \dots$ we find

$$\Delta_0^2 = A_0 + A_1 \cos\left(\frac{\Omega}{2}t\right) + B_1 \sin\left(\frac{\Omega}{2}t\right) \quad (10a)$$

From Eq. (6), $\mathcal{O}(\epsilon)$ dependence terms correspond to

$$\begin{aligned} \frac{d^3}{dt^3} \Delta_1^2 + \frac{\Omega^2}{4} \frac{d}{dt} \Delta_1^2 \\ = -2\delta_1 \Omega \frac{d}{dt} \Delta_0^2 - \frac{\Omega^2}{4} \cos(\Omega t) \frac{d}{dt} \Delta_0^2 + \frac{\Omega^3}{8} \sin(\Omega t) \Delta_0^2 \\ = \delta_1 \Omega^2 \left(A_1 \sin\left(\frac{\Omega}{2}t\right) - B_1 \cos\left(\frac{\Omega}{2}t\right) \right) \\ + A_0 \frac{\Omega^3}{8} \sin(\Omega t) + A_1 \frac{\Omega^3}{8} \sin\left(\frac{3\Omega t}{2}\right) - B_1 \frac{\Omega^3}{8} \cos\left(\frac{3\Omega t}{2}\right) \end{aligned} \quad (10b)$$

In order that a finite answer for Δ_1^2 can be obtained, the resonating terms (frequency $\frac{\Omega}{2}$) have to be removed from the right hand side. This forces $\delta_1 = 0$, both for solution which are like $\cos(\frac{\Omega t}{2})$ and like $\sin(\frac{\Omega t}{2})$. At $\mathcal{O}(\epsilon^2)$, one gets the same value of δ_2 for $\sin(\frac{\Omega t}{2})$ and $\cos(\frac{\Omega t}{2})$ like solutions and hence the unbounded region (the part within two periodic curves) does not exist. This persists at higher order and consequently we do not find any divergence of the width around $\omega = \frac{\Omega}{4}$.

Turning to the region near $\omega = \frac{\Omega}{2}$ we expect a secondary resonance i.e. a width of $\mathcal{O}(\epsilon^2)$ for the divergent region. However, the perturbation technique that we exhibited above does not work in this case and we have to resort to a Krylov–Bogoliubov scheme [16] with the trial solution

$$\begin{aligned} \Delta^2(t) = A_0(t) + A_1(t) \cos(\Omega t) + B_1(t) \sin(\Omega t) \\ + A_2(t) \cos(2\Omega t) + B_2(t) \sin(2\Omega t) \end{aligned} \quad (11)$$

where A_0 , $A_{1,2}$ and $B_{1,2}$ are slowly varying functions of time, so that the $\dot{A}_0$, $\dot{A}_{1,2}$ and $\dot{B}_{1,2}$ terms are negligible. One eventually gets the boundary $\Delta\omega_{1,2}(\epsilon) = \pm \frac{2\epsilon\Omega}{8}$, identical to the boundary for the divergent solution for $\langle x(t) \rangle$ in the vicinity of $\omega = \frac{\Omega}{2}$. This is the reason why no divergence of the variance is seen when the motion of the centre is bounded. Conversely, whenever the particle leaves the trap, its variance also becomes extremely big. This unexpected equality of the boundaries for divergent mean and divergent variance had not been noted before. The boundaries obtained from the numerical analysis of Eq. (6) are shown in Fig. 2, confirming our perturbative solutions.

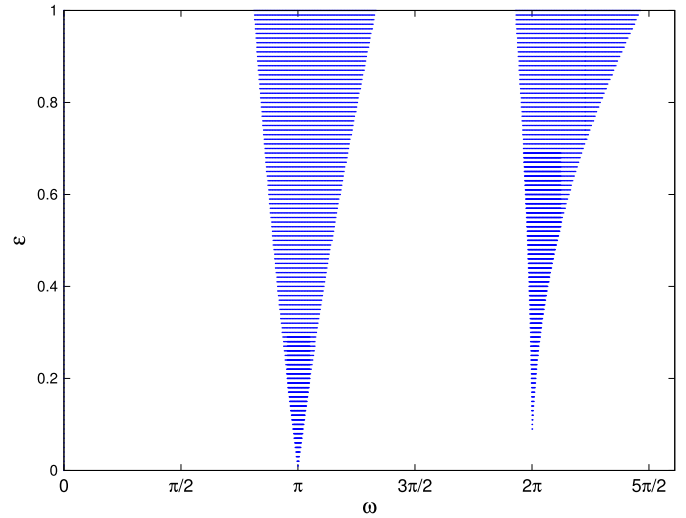


Fig. 2. Zones of bounded and unbounded solutions for Δ^2 . The blue shaded regions show the unstable zones.

We next address the issue whether an initially symmetric wave packet can develop an asymmetry. For a wave packet which is symmetric about the mean $\langle x \rangle$ we must have $\langle (x - \langle x \rangle)^3 \rangle = 0$. Turning to Eq. (7), we note that, if the wave packet is symmetric at $t = 0$, then $S_0 = 0$. Using commutation relations $\dot{S}_0 = [3\langle x^2 p + p x^2 \rangle_0 + 6\langle x \rangle_0^2 \langle p \rangle_0 - 6\langle x p + p x \rangle_0 \langle x \rangle_0] / 2m$, which is also zero for a symmetric packet. Continuing in the same style we find that $\dot{S}_0$ and $\ddot{S}_0$ also vanish and hence it follows that an initial symmetric wave packet will always remain a symmetric wave packet. Returning to Eq. (7), we now ask the question how would an initial asymmetry in the wave function (i.e. $S_0 \neq 0$) time develop. In particular, one needs to know the zones around which the solution of Eq. (7) will wander off to infinity. We use a perturbation expansion similar to that shown in Eqs. (10a) and (10b). Setting $\epsilon = 0$, we note that the zeroth order solution is

$$\begin{aligned} S_0(t) = A_1 \cos(\omega t) + B_1 \sin(\omega t) \\ + A_2 \cos(3\omega t) + B_2 \sin(3\omega t) \end{aligned} \quad (12)$$

The expected primary resonance are then at $\omega = \frac{\Omega}{2}$ and $\frac{\Omega}{6}$. If we explore the region around $\frac{\Omega}{6}$, we find the vanishing of the instability zone exactly as shown in Eq. (10b). Exploring the region around $\frac{\Omega}{2}$, we get the boundary lines $\omega = \frac{\Omega}{2} + \Delta\omega_{1,2}(\epsilon)$ with $\Delta\omega_{1,2}(\epsilon) = \pm \frac{\epsilon\Omega}{8}$ exactly as found for the Mathieu equation shown in Eq. (2). Secondary resonance occurs at $\omega = \Omega, \frac{3\Omega}{2}, 2\Omega, \dots$. Our numerical results for the zones of diverging solutions are shown in Fig. 3.

The kurtosis, K , presents a different story. Before discussing that we point out that an initial Gaussian packet will time develop with the kurtosis unchanged. We note that for a Gaussian wave packet $\langle (x - \langle x \rangle)^4 \rangle = 3[\langle (x - \langle x \rangle)^2 \rangle]^2$ and hence $K = K_G = 3[\langle (x - \langle x \rangle)^2 \rangle]^2$ for such a packet. If we consider $K' = K - K_G$, then it is easy to show that K' satisfies the same equation as K i.e. Eq. (8). We now note that a similar argument, as given for S' , holds for K' . At $t = 0$, if we have a Gaussian wave packet, then $K'(t = 0)$ and all subsequent derivatives $\dot{K}'(t = 0) = \ddot{K}'(t = 0) = \ddot{\ddot{K}}'(t = 0) = \ddot{\ddot{\ddot{K}}}'(t = 0) = 0$, which ensure that the time development of K' will be such that $K' = 0$ always. Hence an initial Gaussian wave packet remains Gaussian forever.

We turn to Eq. (8) and at the zeroth order ($\epsilon = 0$) have the fifth order differential equation

$$\frac{d^5}{dt^5} K_0 + 20\omega^2 \frac{d^3}{dt^3} K_0 + 64\omega^4 \frac{d}{dt} K_0 = 0 \quad (13)$$

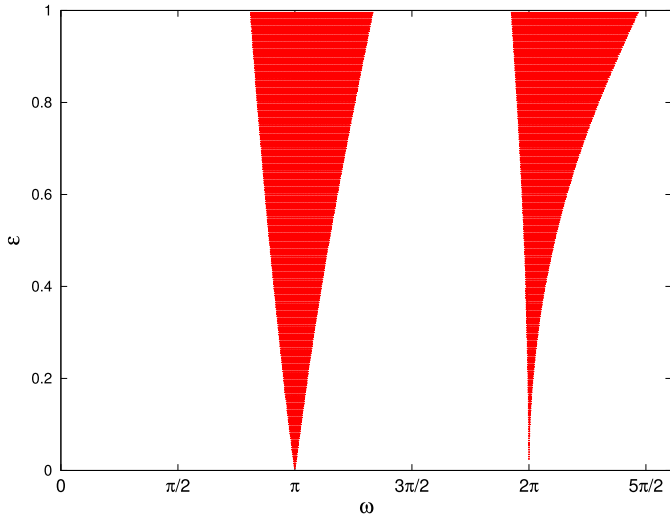


Fig. 3. Zones of bounded and unbounded solutions for S . The red shaded regions show the unstable zones.

with the solution

$$K_0(t) = A_0 + A_2 \cos(2\omega t) + B_2 \sin(2\omega t) + A_4 \cos(4\omega t) + B_4 \sin(4\omega t) \quad (14)$$

Based on this zeroth order answer, we expect primary resonances at $\omega = \frac{\Omega}{8}$ and $\frac{\Omega}{4}$. Focusing on the lowest frequency, we set $\omega = \frac{\Omega}{8} + \Delta\omega_{1,2}(\epsilon)$, with $\Delta\omega = \omega_1\epsilon + \omega_2\epsilon^2 + \dots$ and to $\mathcal{O}(\epsilon)$ obtain,

$$\begin{aligned} & \frac{d^5}{dt^5} K_0 + \frac{5\Omega^2}{16} \frac{d^3}{dt^3} K_0 + \frac{\Omega^4}{64} \frac{d}{dt} K_0 + 5\Omega\omega_1 \epsilon \frac{d^3}{dt^3} K_0 \\ & \frac{\Omega^3 \epsilon \omega_1}{2} \frac{d}{dt} K_0 + \epsilon \left(\frac{d^5}{dt^5} K_1 + \frac{5\Omega^2}{16} \frac{d^3}{dt^3} K_1 + \frac{\Omega^4}{64} \frac{d}{dt} K_1 \right) \\ & + \frac{20\Omega^2 \epsilon \cos(\Omega t)}{64} \frac{d^3}{dt^3} K_0 - \frac{24\Omega^3 \epsilon \sin(\Omega t)}{64} \frac{d^2}{dt^2} K_0 \\ & + \epsilon \left(\frac{128\Omega^4 \cos(\Omega t)}{64 \cdot 64} - \frac{18\Omega^4 \cos(\Omega t)}{64} \right) \frac{d}{dt} K_0 \\ & \epsilon \left(-\frac{\Omega^5 \sin(\Omega t)}{64} + \frac{\Omega^5 \sin(\Omega t)}{16} \right) K_0 = 0 \end{aligned}$$

At the leading order, $\frac{d^5}{dt^5} K_0 + \frac{5\Omega^2}{16} \frac{d^3}{dt^3} K_0 + \frac{\Omega^4}{64} \frac{d}{dt} K_0 = 0$ leading to

$$K_0(t) = A_0 + A_2 \cos\left(\frac{\Omega t}{4}\right) + B_2 \sin\left(\frac{\Omega t}{4}\right) + A_4 \cos\left(\frac{\Omega t}{2}\right) + B_4 \sin\left(\frac{\Omega t}{2}\right) \quad (15)$$

At $\mathcal{O}(\epsilon)$, we have

$$\begin{aligned} & \frac{d^5}{dt^5} K_1 + \frac{5\Omega^2}{16} \frac{d^3}{dt^3} K_1 + \frac{\Omega^4}{64} \frac{d}{dt} K_1 \\ & = -5\Omega\omega_1 \frac{d^3}{dt^3} K_0 - \frac{\Omega^3 \omega_1}{2} \frac{d}{dt} K_0 \\ & - \frac{5\Omega^2 \cos(\Omega t)}{16} \frac{d^3}{dt^3} K_0 + \frac{3\Omega^3 \sin(\Omega t)}{8} \frac{d^2}{dt^2} K_0 \\ & + \frac{\Omega^4 \cos(\Omega t)}{4} \frac{d}{dt} K_0 + \frac{3\Omega^5 \sin(\Omega t)}{64} K_0 \end{aligned} \quad (16)$$

inserting K_0 , $\dot{K}_0$, $\ddot{K}_0$ and $\ddot{K}_0$ from Eq. (15) into the right hand side of Eq. (16), we find (showing explicitly only the resonating terms)

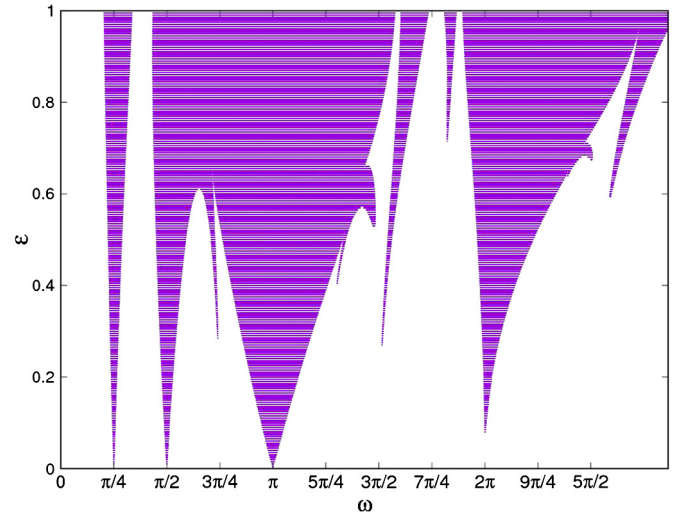


Fig. 4. Zones of bounded and unbounded solutions for K . The violet shaded regions show the unstable zones.

$$\begin{aligned} & \frac{d^5}{dt^5} K_1 + \frac{5\Omega^2}{16} \frac{d^3}{dt^3} K_1 + \frac{\Omega^4}{64} \frac{d}{dt} K_1 \\ & = -\frac{3\Omega^4 \omega_1}{8} A_4 \sin\left(\frac{\Omega t}{2}\right) + \frac{3\Omega^5 \epsilon}{256} A_4 \sin\left(\frac{\Omega t}{2}\right) \\ & + \frac{3\Omega^4 \omega_1}{8} B_4 \cos\left(\frac{\Omega t}{2}\right) + \frac{3\Omega^5 \epsilon}{256} B_4 \cos\left(\frac{\Omega t}{2}\right) \\ & + \text{terms in } \mathcal{O}\left(\frac{3\Omega t}{4}\right), \mathcal{O}\left(\frac{3\Omega t}{2}\right) \text{ etc.} \end{aligned} \quad (17)$$

For the solution $K_1(t)$ to be finite we need to remove the resonating terms from the r.h.s of Eq. (17) and this can be done through the choice:

$$\begin{aligned} \omega_1 &= +\frac{\epsilon\Omega}{32} \quad \text{for a solution } A_4 \sin\left(\frac{\Omega t}{2}\right) \\ \omega_2 &= -\frac{\epsilon\Omega}{32} \quad \text{for a solution } B_4 \cos\left(\frac{\Omega t}{2}\right) \end{aligned} \quad (18)$$

Between the curves $\omega = \frac{\Omega}{8} + \frac{\epsilon\Omega}{32} + \mathcal{O}(\epsilon^2)$ and $\omega = \frac{\Omega}{8} - \frac{\epsilon\Omega}{32} + \mathcal{O}(\epsilon^2)$, we have the divergent zone where the kurtosis blows up in time. In a similar fashion we can explore the region around $\omega = \frac{\Omega}{4}$ and setting $\omega = \frac{\Omega}{4} + \delta$, we arrive at $\omega_1 = \pm \frac{\epsilon\Omega}{16} + \mathcal{O}(\epsilon^2)$ as the two boundaries between which lies the region where the kurtosis blows up. In both these zones, we have the mean position and the width bounded. This leads to the interesting result that for an initially non-Gaussian wave packet, the time development increases the kurtosis indefinitely and at large times, it becomes infinitely large. Hence the indications are that an initially exponentially decaying wave function can change into an algebraically decaying one after a long time. In the region around $\omega = \frac{\Omega}{2}$, which is a secondary resonance, the above approach breaks down and we have to resort to the Krylov–Bogoliubov technique and as in the case of Δ^2 , this again leads to an unusually wide resonance region. The numerical plot of instability zones are shown in Fig. 4. As with the case of $\langle x \rangle$, Δ^2 and S we find the width of the unbound region to be $\mathcal{O}(\epsilon)$ around $\omega = \frac{\Omega}{2}$. However, in this case, the region is wider than for Δ^2 , S and $\langle x \rangle$. Hence even around $\omega = \frac{\Omega}{2}$ there exists zones where Δ^2 and $\langle x \rangle$ are bounded but K is unbounded (see Fig. 5).

We discuss briefly the phase of the wave function. Since $\langle p \rangle = m \frac{d}{dt} \langle x \rangle$, it follows that $\langle p \rangle$ shows a quasi-periodic oscillation in the region where the centre of the wave packet executes bounded motion and increases indefinitely where the classical trajectory escapes from the trap. Referring to the expressions for $k(t)$ and $\gamma(t)$

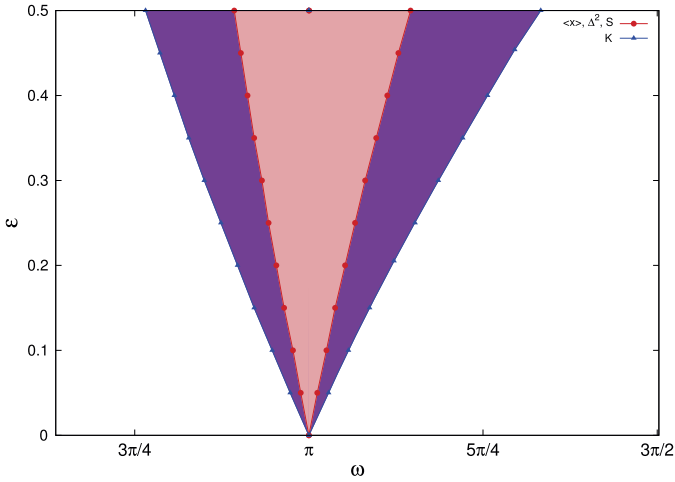


Fig. 5. Comparison of the unstable zones of $\langle x \rangle$, Δ^2 , S and K around $\omega = \frac{\Omega}{2}$. Note that in the violet shaded region all moments except K are bounded.

that we have provided below Eq. (8), we note that $k(t)$ follows exactly the behaviour of $\langle p \rangle$. The function $\gamma(t)$ on the other hand is proportional to the time derivative of $\ln(\Delta^2)$. In the zones where $\langle x \rangle$ and hence also Δ^2 diverge, the behaviour of Δ^2 is proportional to $\exp(t)$ and hence the time derivative of $\ln(\Delta^2)$ is constant. Thus $\gamma(t)$ has oscillatory behaviour when $\langle x \rangle$ is bounded and is virtually time independent when $\langle x \rangle$ is unbounded.

As a final item, we discuss the dynamics of the width Δ_p^2 in momentum space. Straight forward use of Ehrenfest's relation leads to the dynamics

$$\begin{aligned} \frac{d^3}{dt^3} \Delta_p^2 + 4\omega^2(1 + \epsilon f) \frac{d}{dt} \Delta_p^2 &= \frac{3\epsilon \dot{f}}{1 + \epsilon f} \frac{d^2}{dt^2} \Delta_p^2 \\ + \left(\frac{(1 + \epsilon f)(\epsilon \ddot{f}) - 3(\epsilon \dot{f})^2}{(1 + \epsilon f)^2} \right) \frac{d}{dt} \Delta_p^2 &+ 2\omega^2 \epsilon \dot{f} \Delta_p^2 \end{aligned} \quad (19)$$

As in the case of the variance in co-ordinate space, Δ_p^2 does not have any instability near $\omega = \frac{\Omega}{2}$. The instability zones are exactly where the zones of Δ^2 are and incidentally are of the same width. This is a remarkable result when we consider the big difference is the corresponding differential equations [Eqs. (7) and (19)].

A possible extension of this work could be the question of coupling the system that we have studied to an external heat bath and

looking for the corresponding quantum Langevin equation. Recent work in this direction for the more complicated double well (both symmetric and anti-symmetric) potentials have been carried out by Magazzù et al. ([17], [18]) and Valenti et al. [19]. A particularly interesting result is the stabilization of a metastable state against quantum fluctuations due to the dissipative coupling with the heat bath. Studying such a system from the Ehrenfest point of view should be an interesting avenue.

In conclusion, we have used Ehrenfest's theorem to find the time development of the first few moments of the probability distribution of a quantum particle moving in a Paul trap. All the dynamical equations are linear ordinary differential equation and convenient to analyze. The most interesting qualitative result is that for *non-Gaussian* initial wave packets the *centre*, the *variance*, and *skewness* become infinitely large with time in the same regions of the $(\omega - \epsilon)$ space but the *kurtosis* diverges in a wider region, which includes the instability regions of $\langle x \rangle$, Δ^2 , and S . This would imply that lower moments can remain finite also in the presence of a drastic change in the tail of the probability distribution as time develops. The increased kurtosis implies that there will be vastly increased probability of finding the particle far from the centre even though its motion is bounded.

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